MIL-C-915C
15 March 1968
SUPERSEDING
MIL-C-915B(NAVY)
23 April 1965
(See 6.7)

MILITARY SPECIFICATION

CABLE, ELECTRICAL, SPECIAL PURPOSE,

GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers special purpose electrical cable.

1.2 Classification. Cable covered by this specification shall be of the types, sizes, and construction specified on the specification sheets.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip (Polyolefin).
QQ-R-175	- Resistance Wire.
TT-P-320	- Pigment, Aluminum; Powder and Paste, For Paint.
UU-P-271	- Paper, Wrapping, Waterproofed Kraft.
PPP-V-205	- Veneer, Paper-Overlaid, Container Grade.

MILITARY

MIL-G-160	- Grips, Cable (Naval Shipboard Use).
MIL-C-572	- Cords, Yarns, and Monofilaments, Organic Synthetic Fiber.
MIL-C-12000	- Cable, Cord, and Wire, Electric, Packaging of.
MIL-L-17331	- Lubricating Oil, Steam Turbine (Non-Corrosive).
MIL-I-45208	- Inspection System Requirements.

See Supplement-1 for applicable specification sheets.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

MILITARY

815-F-1197030 - Tubes, Stuffing, Steel, for Passing Electric Cables Through Pressure-proof BHDS, Submarines.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-B3 - Soft or Annealed Copper Wire.
ASTM-B8 - Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft.
ASTM-B33 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.
ASTM-B172 - Rope-Lay Stranded Copper Conductors Having Bunch-Stranded Members, for Electrical Conductors.
ASTM-B173 - Rope-Lay Stranded Copper Conductors Having Concentric Members, for Electrical Conductors.
ASTM-B174 - Bunch-Stranded Copper Conductors for Electrical Conductors.
ASTM-B189 - Lead-Coated and Lead Alloy-Coated Soft Copper Wire for Electrical Purposes.
ASTM-B193 - Resistivity of Electrical Conductor Materials.
ASTM-B228 - Concentric-Lay-Stranded Copper Clad Steel Conductors.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNDERWRITERS' LABORATORIES, INC.

(Application for copies of publications should be addressed to Underwriters' Laboratories, Inc. 161 Sixth Avenue, New York, N. Y. or 207 East Ohio Street, Chicago, Illinois.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. Detail requirements or exceptions to the general requirements specified herein shall be as specified on the specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern.

3.2 Qualification (when specified on the specification sheet). When specified on the specification sheet, cables furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.5).

3.3 Materials. All materials used in the construction of those cables requiring qualification shall have prior written approval of the qualifying activity.

3.3.1 Conductors.

3.3.1.1 Standard copper conductors. Standard copper conductors shall be composed of soft or annealed copper strands conforming to ASTM-B3.

3.3.1.2 High strength conductors. High strength conductors shall be composed of 30-percent conductivity, high strength, copper-covered steel strands conforming to ASTM-B228.

3.3.1.3 Coating. Conductor coating shall be commercially pure tin in accordance with ASTM-B33 or alloy coated in accordance with ASTM-B189.

3.3.2 Insulation. The type of insulation material and its physical and electrical properties shall be as required by the applicable specification sheet.

3.3.3 Jackets. The material used for the jackets over conductors and groups of conductors, and for the cable jacket, shall have the physical and electrical properties specified on the applicable specification sheet.

3.3.4 Shields. Unless otherwise specified on the applicable specification sheet, shields shall consist of braided, tin-coated, copper wires, A.W.G. No. 34 or 36.

3.3.5 Fillers. Unless otherwise specified on the applicable specification sheet, fillers shall consist of cotton, jute, synthetic fibers, glass fibers, or asbestos. The filler material selected shall be appropriate for the type of cable in which it is used. Fillers shall be treated for flame and moisture resistance, as necessary.

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3.3.6 Tape. Tape shall be as specified on the applicable specification sheet.

3.3.7 Braids (marker). Marker braids shall be of rayon conforming to MIL-C-572.

3.3.8 Tie cord. Tie cords shall be of cotton, glass, or synthetic fiber having a maximum diameter of 0.065 inch and a minimum breaking strength of 30 pounds. All tie cords shall be treated with a flexible waterproofing compound.

3.3.9 Separators. Unless otherwise specified on the specification sheet, separators, when required, shall consist of a wind of glass fibers, synthetic fibers, or cotton; they may also be a tape of cotton, synthetic fiber, paper, or polyester. All separators shall be applied spirally and in such a manner so as to provide full conductor coverage during manufacture and use of the cable. The choice of separator material shall be appropriate for the type of cable in which it is employed.

3.3.10 Binders. Binders shall be of the material specified on the applicable specification sheet, or shall be a type approved by the qualifying activity for the specific cable construction.

3.3.11 Reinforcement. Reinforcing binders or reinforcement for a single or double layer jacket shall be size 3A nylon cord or an alternate approved by the qualifying activity.

3.3.12 Armor. The armor shall consist of Alclad, 5056 aluminum alloy wires having a diameter of 0.0126 ± 0.0005 inch, a minimum tensile strength of 50,000 pounds per square inch (p.s.i.), and a minimum elongation (before application to the cable) of 2 percent.

3.4 Construction.

3.4.1 Conductor stranding. Conductors shall be concentric-lay-stranded as shown in table I or bunch- and rope-lay-stranded as shown in table II, as required by the applicable specification sheet.

3.4.1.1 Concentric-lay-stranded. The concentric-lay-stranded conductors shall be in accordance with ASTM-B8.

3.4.1.2 Bunch-stranded. The bunch-stranded conductors shall be in accordance with ASTM-B174.

3.4.1.3 Rope-lay-stranded. The rope-lay-stranded conductors shall be in accordance with ASTM-B172 or ASTM-B173.

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Table I — Conductor data, concentric strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Max. conductor resistance (d. c.) per 1000 feet at 25°C		Weight per 1000 feet
	Minimum	Nominal (Inch)	Nominal (Inches)	Nominal (Cir. mils)	Minimum (Cir. mils)	Bare (Ohms)	Coated (Ohms)	Approx. (Pounds)
1(7)	7	0.013	0.038	1,119	1,096	9.89	10.3	3.4
2(7)	7	.016	.048	1,779	1,743	6.25	6.50	5.5
3(7)	7	.020	.060	2,828	2,771	3.92	4.07	8.7
3(19)	19	.013	.063	3,036	2,975	3.64	3.80	9.3
4(7)	7	.025	.076	4,497	4,407	2.46	2.52	14
6(7)	7	.031	.092	6,512	6,382	1.69	1.73	20
9(7)	7	.036	.108	9,016	8,836	1.23	1.25	28
14(7)	7	.045	.136	14,340	14,050	0.770	0.784	44
23(7)	7	.057	.171	22,800	22,340	.486	.493	70
30(19)	19	.040	.202	30,860	30,240	.358	.365	95
40(19)	19	.045	.226	38,910	38,130	.284	.288	120
50(19)	19	.051	.254	49,080	48,100	.225	.228	150
60(37)	37	.040	.282	60,090	58,890	.185	.189	190
75(37)	37	.045	.317	75,780	74,260	.146	.149	230
100(61)	61	.040	.363	99,060	97,080	.112	.115	310
125(61)	61	.045	.407	124,900	122,400	.0888	.0904	390
150(61)	61	.051	.457	157,600	154,400	.0704	.0716	490
200(61)	61	.057	.514	198,700	194,700	.0560	.0570	610
250(61)	61	.064	.577	250,500	245,500	.0444	.0453	770
300(91)	91	.057	.628	296,400	290,500	.0375	.0382	910
350(91)	91	.062	.682	349,800	342,800	.0316	.0321	1,100
400(127)	127	.057	.742	413,600	405,400	.0268	.0273	1,300
500(127)	127	.064	.832	521,600	511,100	.0214	.0217	1,600
650(127)	127	.072	.936	657,600	644,500	.0169	.0172	2,000
800(127)	127	.081	1.050	829,300	812,700	.0134	.0136	2,600
1000(127)	127	.091	1.180	1,046,000	1,025,000	.0106	.0108	3,200
1300(127)	127	.102	1.325	1,318,000	1,292,000	.00843	.00851	4,100
1600(127)	127	.114	1.485	1,662,000	1,629,000	.00668	.00676	5,100
2000(127)	127	.128	1.670	2,097,000	2,055,000	.00530	.00536	6,300

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Table II — Conductor data, bunch and rope-lay strand.

Standard copper conductor size	Number of strands	Strand diameter	Diameter over conductor	Cross sectional area of conductor		Max. conductor resistance (d. c.) per 1000 feet at 25°C.		Weight per 1000 feet
	Minimum	Nominal (Inch)	Nominal (Inches)	Nominal (Cir. mils)	Minimum (Cir. mils)	Bare (Ohms)	Coated (Ohms)	Approx. (Pounds)
Bunch:								
1/2(21)	21	0.005	0.028	525	515	21.1	22.2	1.6
3/5(7)	7	.010	.030	703	689	15.7	16.4	2.2
1(10)	10	.010	.038	1,005	985	11.0	11.5	3.1
1(26)	26	.006	.042	1,034	1,014	10.7	11.2	3.2
1-1/2(16)	16	.010	.049	1,608	1,576	6.88	7.17	4.9
1-1/2(41)	41	.006	.049	1,630	1,597	6.78	7.15	5.0
2-1/2(19)	19	.011	.057	2,426	2,300	4.75	4.90	6.3
2-1/2(26)	26	.010	.061	2,613	2,561	4.23	4.48	8.0
2-1/2(65)	65	.006	.061	2,594	2,542	4.26	4.51	8.0
4(41)	41	.010	.077	4,121	4,038	2.68	2.81	13
6(65)	65	.010	.097	6,533	6,402	1.69	1.78	20
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope:								
20(49)	7x7	.020	.180	19,800	19,400	.562	.582	61
23(228)	19x12	.010	.190	22,910	22,460	.499	.523	73
26(49)	7x7	.023	.210	26,250	25,730	.428	.440	82
30(304)	19x16	.010	.220	30,550	29,940	.374	.392	97
33(336)	7x48	.010	.235	33,370	33,090	.344	.360	106
42(209)	19x11	.014	.260	42,110	41,280	.272	.284	130
42(49)	7x7	.029	.260	41,740	40,910	.268	.275	130
53(532)	19x28	.010	.304	53,470	52,400	.213	.218	169
60(304)	19x16	.014	.310	61,260	60,040	.187	.196	190
66(133)	19x7	.022	.330	66,370	65,040	.171	.175	210
84(2107)	2107	1/	.410	83,690	82,020	.138	------	270
83(418)	19x22	.014	.380	84,230	82,560	.136	.142	270
105(2646)	2646	1/	.460	105,500	103,400	.108	------	340
105(259)	37x7	1/	.410	105,500	103,400	.108	.113	330
133(3325)	3325	1/	.520	133,100	130,400	.0876	------	430
133(259)	37x7	1/	.460	133,100	130,400	.0856	.0892	420
133(684)	19x36	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	19x40	.014	.510	153,100	150,100	.0747	.0780	490
168(427)	61x7	1/	.520	167,800	164,400	.0681	.0710	530
200(988)	19x52	.014	.580	199,100	195,100	.0575	.0600	630
220(259)	37x7	.029	.610	220,700	216,300	.0494	.0507	718
250(1254)	19x66	.014	.680	252,700	247,700	.0453	.0472	800
250(6384)	19x7x48	.006	.713	250,000	245,000	.0461	-------	821
300(259)	37x7	.034	.714	300,000	294,000	.0378	-------	926
400(2052)	19x108	.014	.820	413,500	405,300	.0277	.0289	1,300
500(259)	37x7	.044	.922	500,000	490,000	.0234	-------	1,585
600(427)	61x7	.038	1.013	600,000	588,000	.0196	-------	1,910
650(427)	61x7	.039	1.053	650,000	637,000	.0154	-------	2,070
800(4033)	37x109	.014	1.150	812,700	796,500	.0141	.0148	2,600

1/ These are conductors for flexible cables, for which the minimum number of strands and minimum area are specified. The individual strand diameter is determined from the required area and the actual number of strands used.

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3.4.2 Shields. Shields shall be of braided construction. The angle of application and the percent coverage shall be as required by the applicable specification sheet. The percent coverage and the angle of application shall be determined by the following formula:

$$\text{Percent coverage}(k) = 100(2F-F^2)$$

Where $F = NPd/\text{Sin } a$

$\text{Tan } a = 2\pi DP/C$

a = acute angle of braid with axis of cable
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires per carrier
C = number of carriers
P = picks per inch of cable length

3.4.3 Armor, braided metal. The weave shall be such that the wires of a carrier are laid closely together, flat and parallel. The braid shall be applied with maximum tension possible so as to prevent loosening and creeping but not to cause excessive broken ends. Splices in the braid strands shall be infrequent, staggered and inconspicuous so as not to increase the nominal diameter of the cable or result in rough spots. The braid shall remain snug to the cable jacket and not spring away when cut. It shall provide reasonable mechanical protection to the jacket during installation and handling, and shall retard deformation of the cable under compression from stuffing tubes and cable hangers.

3.4.3.1 Construction of braided metal armor. The weave shall be of either the "one over-one under" or the "two over-two under" type. The selection of the number of ends per carrier and the number of carriers per braider shall be such as to produce a basket weave with a braid angle within the limits shown in table III. The number of ends per carrier shall be not more than as shown in table IV.

Table III - Armor braid angle.

Diameter over sheath (Inches)	Braid angle	
	Minimum (Degrees)	Maximum (Degrees)
0.100 to 0.600	30	60
.601 to 1.000	35	60
1.001 to 1.500	40	70
1.501 to 2.000	45	70
2.001 to 3.000	50	80

Table IV — Ends per carrier.

Cable diameter under braid (Inches)	Maximum number of ends per carrier	
	One over-one under	Two over-two under
0.100 to 0.400	8	5
.401 to .800	12	10
.801 to 3.000	16	10

3.4.3.2 Coverage. The coverage for all types and sizes of cable shall be not less than 88 percent, as determined by the following formula:

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$$\text{Percent coverage}(K) = 100(2F - F^2)$$

Where, $F = NPd/\sin a$

$\tan a = 2\pi DP/C$

a = acute angle of braid with axis of cable
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires in carrier
C = number of carriers
P = picks per inch of cable length

3.4.4 Aluminum paint. Aluminum paint shall consist of aluminum pigment (paste form) conforming to TT-P-320, in a vehicle of the synthetic resinous (phenol-formaldehyde or glyceride) type designed for good penetration, high bonding power, resistance to water and to oil, and with flexibility at temperatures up to 100°C.

3.4.4.1 Application. Painting shall be done as soon as practicable after completion of manufacturing operations so as to prevent accumulation of dirt, moisture, and foreign matter on the braid or in the interstices. The paint shall be properly dried and shall not be tacky. The painting shall be done preferably by the immersion method so as to fill all interstices in the outer cable surface. The paint finish shall provide a relatively smooth overall coating that will facilitate making watertight glands with stuffing tubes and packing. It shall be flexible and adhere firmly so that it will not crack, peel, or flake off under the cold bend test.

3.4.5 Dimensional tolerances. Where minimum or maximum dimensions, or both, are specified, no minus or plus tolerances, respectively, will be permitted. Where no minimum overall cable diameter is specified, the minimum permissible diameter shall be not less than 92-1/2 percent of the specified maximum overall cable diameter.

Note: The maximum overall diameter of unarmored cables shall be 0.050 inch less than the maximum specified for the armored counterpart.

3.4.5.1 Thickness (cable jacket). The average thickness of a cable jacket measured at any cross section shall be not less than the specified nominal. The minimum thickness at any cross section shall be not less than 90 percent of the specified nominal.

3.4.6 Centering and circularity.

3.4.6.1 Insulation. The insulation on the individual conductors shall be uniform in diameter throughout the conductor length. At any cross section the maximum wall thickness shall not exceed the minimum by more than 25 percent for specified thicknesses greater than 0.025 inch, nor by more than 40 percent for specified thicknesses of 0.025 inch and less.

3.4.6.2 Cable jacket. The cable jacket shall be applied concentrically to the cable core in a manner to maintain circularity in the completed cable. The maximum wall thickness of the jacket at any cross section shall not exceed the minimum by more than 66 percent. In case of multi-conductor cables the jacket thickness shall be determined from measurements made at the high point of each conductor taken on a line through the center of the cable and through the center of each conductor in the outer layer.

3.4.7 Manufacturer's identification. All cable shall be provided with a continuous, thin, moisture-resistant marker tape, not less than 1/10 inch wide. The marker tape shall be placed directly under the jacket unless otherwise approved by the qualifying activity. The tape shall be printed to show the following information at intervals no greater than 1 foot: Name and location of manufacturer; year of manufacture; specification number (MIL-C-915); and progressive serial number. The serial number is not necessarily a footage marker. A serial number shall not be repeated by a manufacturer in any one year for any one type and size of cable.

3.4.7.1 Marker threads. Marker threads shall be used only when satisfactory to the qualifying activity. If approved, marker threads shall be in accordance with marker threads assigned by Underwriters' Laboratory, (U.L.) Incorporated, to indicate the manufacturer of the cable. When U.L. marker threads are used, Peerless Marker Threads (or equal), shall also be used to indicate the year of manufacture, as follows:

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Year	Color Code
1967	Brown
1968	Black
1969	Pink
1970	Red-Blue
1971	Red-Yellow
1972	Red-Slate
1973	Red-Green
1974	Red-Brown
1975	Red-Black
1976	Red-Pink

3.4.8 Identification coding. Individual conductors and groups of conductors, such as pairs and triads, shall be identified in accordance with table V. All coding shall remain intact and legible after the conductors or groups are fanned out at the cable ends and, where necessary, wiped clean of any sealing compound. The identification shall not rub off after exposure to hydraulic fluids and handling incident to installation and maintenance.

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Table V — Identification code.

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
1	Black	---	---
2	White	---	---
3	Red	---	---
4	Green	---	---
5	Orange	---	---
6	Blue	---	---
7	White	Black	---
8	Red	Black	---
9	Green	Black	---
10	Orange	Black	---
11	Blue	Black	---
12	Black	White	---
13	Red	White	---
14	Green	White	---
15	Blue	White	---
16	Black	Red	---
17	White	Red	---
18	Orange	Red	---
19	Blue	Red	---
20	Red	Green	---
21	Orange	Green	---
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange

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Table V — Identification code (cont'd.).

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	---	---
39	Brown	Black	---
40	Brown	White	---
41	Brown	Red	---
42	Brown	Green	---
43	Brown	Orange	---
44	Brown	Blue	---
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red
48	Orange	Red	Blue
49	Blue	Red	Orange
50	Black	Orange	Red
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue

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Table V — Identification code (cont'd.).

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange
81	Blue	Black	Green
82	Orange	White	---
83	Green	Red	---
84	Black	Green	---
85	White	Green	---
86	Blue	Green	---
87	Black	Orange	---
88	White	Orange	---
89	Red	Orange	---
90	Green	Orange	---
91	Blue	Orange	---
92	Black	Blue	---
93	White	Blue	---
94	Red	Blue	---
95	Green	Blue	---
96	Orange	Blue	---
97	Yellow	---	---
98	Yellow	Black	---
99	Yellow	White	---
100	Yellow	Red	---
101	Yellow	Green	---
102	Yellow	Orange	---
103	Yellow	Blue	---
104	Black	Yellow	---
105	White	Yellow	---

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Table V — Identification code (cont'd.).

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
106	Red	Yellow	---
107	Green	Yellow	---
108	Orange	Yellow	---
109	Blue	Yellow	---
110	Black	Yellow	Red
111	White	Yellow	Red
112	Green	Yellow	Red
113	Orange	Yellow	Red
114	Blue	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.4.8.1 Conductor coding. The use of colors or printed numbers and words on the individual conductors shall be as specified on the applicable specification sheet.

3.4.8.2 Group coding. Group coding may be accomplished by using marker braids, printed insulating tapes over the group shield, printed binder tapes over the group, or by printing on a group jacket. The method used shall be as specified on the applicable specification sheet.

3.4.8.3 Printed identification. Printed coding shall consist of the appropriate color number followed by the corresponding spelled-out word or words. The legend shall be printed in contrasting color; preferably white ink on a black or dark background and black ink on a white or light background. The legend shall be repeated at 3-inch maximum intervals and alternate legends shall be inverted. For example: 10-ORANGE-BLACK ꓘCAꓤ-ƎGNAЯO-0I. The print shall be block or italic type with a minimum height of 3/64 inch.

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3.4.8.4 Marker braids. Marker braids shall be closely woven rayon. The braids shall be treated to prevent fraying at the cut ends incident to handling during installation.

3.4.8.5 Telephone cable coding. Color coding for telephone cables shall be accomplished by using the following solid colors: (1) black, (2) white, (3) red, (4) green, (5) orange, (6) blue, (7) brown, (8) gray, (9) yellow, (10) purple, (11) tan and (12) pink. The colors shall be combined to form pairs as follows:

(a) (1) Black paired with 2 through 12 for first 11 pairs.
(b) (2) White paired with 3 through 12 for next 10 pairs.
(c) (3) Red paired with 4 through 12 for next 9 pairs.
(d) (4) Green paired with 5 through 12 for next 8 pairs.
(e) (5) Orange paired with 6 through 12 for next 7 pairs.
(f) (6) Blue paired with 7 through 12 for next 6 pairs.
(g) (7) Brown paired with 8 through 12 for next 5 pairs.
(h) (8) Gray paired with 9 through 12 for next 4 pairs.
(i) (9) Yellow paired with 10 through 12 for next 3 pairs.
(j) (10) Purple paired with 11 through 12 for next 2 pairs.
(k) (11) Tan paired with 12.

3.4.8.6 Special color code. When special color coding is required, the following sequence of solid colors shall be used: (1) black, (2) white, (3) red, (4) green, (5) orange, (6) blue, (7) brown, (8) gray, (9) yellow, (10) purple, (11) tan and (12) pink.

3.4.9 End seals. All types and sizes of cable specified herein shall have both ends of every shipping length provided with a seal to prevent the entrance of moisture. The method of sealing cable ends shall be subject to approval of the qualifying activity.

3.4.10 Surface marking. When surface marking of completed cable is required by the applicable specification sheet, ink marking shall be used. The marking shall be legible, durable, and shall not indent, deform, or otherwise damage the jacket. The legend shall be repeated at 2-foot maximum intervals, and shall contain the manufacturer's name and location and cable type designation.

#. 3.4.11 Watertightness. When a watertight cable is specified on the applicable specification sheet, all voids within the cable construction shall be filled with a sealing compound to prevent the passage of water longitudinally through the cable. The sealing compound shall be compatible with all of the cable components. It shall not harden during service or storage of the cable, and it shall be readily removable without damage to other cable components when making connections.

3.4.12 Jacket reinforcement. When a reinforced polychloroprene jacket is required by the applicable specification sheet, the jacket shall be applied in two layers with a size 3A nylon cord applied as a braid or two reverse serves between the layers. The minimum thickness of the outer layer of jacket shall be 50 percent of the total. The reinforcement shall be applied with approximately four picks or crossovers per inch.

3.5 Electrical properties. All electrical properties of the completed cable shall be as required by the applicable specification sheet.

3.6 Physical properties. All physical properties of the completed cable and cable components shall be as required by the applicable specification sheet.

3.7 Repair of insulation or cable jacket. Repair of cable jacket will not be permitted on unarmored cables which are required to pass a hydrostatic test. Repair of cable jacket will be permitted on cables which are not required to pass a hydrostatic test, provided the materials and techniques used are such that the finished cable complies with all the requirements of this specification. The materials and techniques used in the repair of either insulation or cable jacket shall be subject to approval of the qualifying activity. The frequency of repairs shall be held to a minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable

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for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall have an inspection system which meets the requirements specified in MIL-I-45208. The inspection system shall include the establishment and maintenance of in-process inspections to be performed at appropriate locations within the supplier's plant for the purpose of assuring continuous control of the manufacture of components and assemblies.

4.2 Classification of inspection. The examination and testing of all cable covered by this specification shall be classified as follows:

(a) Qualification inspection.
(b) Quality conformance inspection.
  (1) Group A examination and tests.
  (2) Group B tests.
  (3) Group C tests.
(c) Verification inspection.
(d) Comparison inspection.
(e) Conductor inspection.

4.2.1 The test applicable to each classification shall be as shown in table VI.

Table VI — Categories of inspection.

Examination or test	Test paragraph	Qualification inspection	Quality conformance inspection (Group)	Comparison inspection	Conductor inspection
Conductors inspection	4.7				X
Visual and dimensional	4.8.1	X	A	X	
Physical tests on insulation and jacket					
Original	4.8.2	X	B	X	
Accelerated aging	4.8.2	X	C	X	
Armor	4.8.3	X		X	
Permanence of printing	4.8.4	X	B	X	
Cold bend	4.8.5	X	B	X	
Cold working	4.8.6	X	C		
Drip	4.8.7	X	B	X	
Thermoplastic flow	4.8.8	X			
Watertightness	4.8.9	X	A	X	
Hydrostatic (open end)	4.8.10	X	A	X	
Breaking strength	4.8.11	X	B	X	
Bending endurance	4.8.12	X	C		
Twisting	4.8.13	X	C	X	
Tension (cable)	4.8.14	X	C	X	
Stress endurance	4.8.15	X		X	
Pressure cycling	4.8.16	X			
Flammability	4.8.17	X	C	X	

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Table VI - Categories of inspection (Continued).

Examination or test	Test paragraph	Qualification inspection	Quality conformance inspection	Comparison inspection	Conductor inspection
			(Group)		
Conductor resistance	4.9.1	X	A		
Dielectric strength	4.9.2	X	A		
Insulation resistance	4.9.3	X	A		
Capacitance	4.9.4	X	A	X	
Characteristic impedance	4.9.5	X	A	X	
Pulse response time	4.9.6	X	A		
Attenuation	4.9.7	X	B		
Mutual inductance	4.9.8	X	B	X	

4.3 Qualification inspection.[1/] Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification inspection shall consist of the examination and tests shown in table VI.

4.3.1 Qualification specimens. At least one specimen of cable for which individual type or group qualification is sought shall be submitted for qualification inspection. The cable type, size, and length of specimens required for group qualification shall be in accordance with table VII.

Table VII — Qualification test specimens.

Group	Qualifying type and size	Length of specimen (feet)	Types in group
1	DHFR-4	200	SHFR, DHFR, THFR
2	TTHFWA-10	200	TTHFWA, PBTM, TSP - all sizes
3	MDY-23	200	MDY, MDU - all sizes
4	DSWS-4	200	DSWS
5	DSS-3	200	DSS, TSS, FSS, 7SS - all sizes
6	MCSF-4	200	MCSF
7	MHOF-30	200	MHOF - all sizes
8	TTOP-10	200	TTOP - all sizes
9	MCOS-6	200	MCOS - all sizes
10	MMOP-5	200	MMOP
11	THOF-9	200	DHOF, THOF, FHOF - sizes 9 and smaller
12	THOF-150	200	HOF - sizes 14 and larger, CVSF and T3P - all sizes
13	TTRS-8	200	TTRS, TTRSA - all sizes
14	DLT	200	DLT
15	JAS-250	200	JAS
16	TPU-6	200	TPU-6

[1/] Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". (See 6.5).

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The groups are based on similar characteristics and requirements. The qualifying activity may extend approval of any one type and size of cable to cover any or all types in the same group; however, the qualifying activity may require performance of any or all qualification inspections on additional types in the same group prior to such extension of approval.

4.4 Definition of terms. For the purpose of this section, the following definitions apply:

4.4.1 Inspection lot. An inspection lot shall consist of the total number of units of product of any one type, size, and construction manufactured under essentially the same conditions.

4.4.2 Unit of product. A unit of product is the unit ordering length as specified on the specification sheet, except that when the unit ordering length is 500 feet, two such lengths shall constitute a unit of product. Random and remnant lengths or special order lengths may be added together to equal a unit ordering length.

Note: Length tolerances as given for standard lengths shall apply. (See 6.2.2).

4.4.3 Sample. A sample is one unit of product selected from an inspection lot.

4.4.4 Specimen. A specimen is an individual length of cable or an individual length of a component of cable which has been taken from a sample.

4.4.5 Defective unit. A defective unit is a unit of product from which a specimen that failed to meet one or more requirements was taken.

4.4.6 Rejected lot. A rejected lot is a lot which contains more than the acceptable number of defective units.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed on all completed cable. This inspection shall consist of groups A, B, and C examinations and tests, as applicable. Dielectric strength and insulation resistance tests shall be performed on 100 percent of completed cable. The conductor resistance test shall be performed on 100 percent of conductors which have an area less than 1600 circular mils (nominal); for conductors which have an area greater than 1600 circular mils (nominal), the conductor resistance test shall be performed on conductors selected as follows:

No. of conductors in cable	No. of conductors selected for test
1 to 4	all
5 to 20	4
21 to 30	5
31 to 42	6
43 to 56	7
57 to 72	8
73 to 91	9
92 to 110	10
111 to 132	11
133 to 156	12

Except for the dielectric strength, insulation resistance, and conductor resistance tests, groups A, B, and C examination and tests shall be performed on samples.

4.5.1 Sampling procedure. The required number of samples shall be selected at random from the inspection lot. All nonconforming starting and finishing ends of the completed cable shall be removed by the supplier prior to selecting samples.

4.5.1.1 Sampling and accept/reject criteria for group A examination and tests. Samples for group A examination and tests shall be selected in accordance with the first two columns of table VIII.

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Table VIII — Sampling for group A examination and tests.

Number of units of product in lot	Number of samples	Acceptance number defective units	Rejection number defective units
10 or under	2	0	1
11 to 30	3	0	1
31 to 90	7	0	1
91 to 210	15	1	2
211 and over	25	2	3

4.5.1.2 Sampling for group B tests. Samples for group B tests shall be selected in accordance with table IX.

Table IX — Sampling for group B tests.

Units of product	Number of samples
8 and under	1
9 to 30	2
31 to 90	3
91 to 210	4
211 and over	5

4.5.1.3 Sampling for group C tests. Samples for group C tests shall be selected in accordance with table X.

Table X — Sampling for group C tests.

Two months' production (units of product)	Number of samples
8 or under	1
9 to 30	2
31 to 90	3
91 to 210	4
211 and over	5

4.5.1.4 Tightened sampling and accept-reject criteria. If the number of defective units in a lot exceeds the acceptance number for the group A tests (see table VIII) or if there are any failures in group B tests, the lot shall be held in abeyance. The manufacturer shall have the option of reworking the lot and resubmitting such a lot for examination and testing on a tightened sampling basis in accordance with table XI. Any failure in group C tests or tests performed on tightened sampling basis shall be cause for rejection of the lot.

Table XI — Tightened sampling.

Number of units of product in lot	Number of samples
10 or under	3
11 to 30	5
31 to 90	9
91 to 210	15

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4.5.1.5 Action in case of rejection. Any completed cable which does not comply with specifications, cannot be made to comply, and is therefore not acceptable, shall be recorded with the Government inspector. The record shall be furnished within 10 days after rejection, and shall show the type, size, lengths, and serial numbers of the marker tape recorded from each end of each length.

4.5.2 Group A examination and tests. The applicable examination and tests from table VI shall be performed on specimens taken from samples selected in accordance with table VIII.

4.5.3 Group B tests. The applicable group B tests from table VI shall be performed on specimens taken from samples selected in accordance with table IX.

Note 1: The number of conductors per sample from which specimens shall be taken shall be as follows:

Number of conductors in cable	Number of conductors for specimens
1 to 4	all
5 to 20	4
21 to 30	5
31 to 42	6
43 to 56	7
57 to 72	8
73 to 91	9
92 to 110	10
111 to 132	11
133 to 156	12

Note 2: Five specimens of insulation from each conductor chosen and five specimens of jacket from each sample shall be tested. The test result shall be the median value obtained from each five jacket or insulation specimens.

4.5.4 Group C tests. The group C tests shown in table VI shall be performed on specimens taken from group C samples selected in accordance with table X. (See notes 1 and 2 of 4.5.3.)

4.6 Verification and comparison inspection. To insure that manufacturers continue to supply cable which conforms to specification requirements, verification and comparison inspection, in addition to inspection normally conducted by a Government inspector, shall be performed on all cable requiring Government source inspection at the manufacturer's plant.

4.6.1 Verification inspection. Verification inspection shall be conducted at the manufacturer's plant under the surveillance of a Government representative. The inspection shall consist of the applicable tests of groups A, B, and C as required for quality conformance inspection. The sample used and test results obtained from it may be used as a part of the quality conformance inspection for that particular lot.

4.6.1.1 Sampling procedure for verification inspection. One sample shall be selected from every fifth lot, as a minimum, of each type and size of cable for which qualification is required.

4.6.2 Comparison inspection. Comparison inspection shall be conducted outside the manufacturer's plant, at a Government-designated laboratory. The inspection shall consist of the applicable comparison inspection tests listed in table VI.

4.6.2.1 Sampling procedure for comparison inspection. One sample shall be selected by a Government representative for comparison inspection from the first lot of each type and size of cable produced by a manufacturer after his product has been qualified. Thenceforth, one sample of the type and size required for qualification shall be selected every 4 months, or as near as possible to every 4 months. If the specific type and size is not available for sampling at the proper time, a sample from the lot of the nearest type and size that falls within a qualifying group shall be used. The entire lot of cable represented by the sample

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submitted for comparison inspection shall be retained by the manufacturer until notice of the inspection results are received from the Government.

4.6.2.2 Length of specimens. The length of specimens required for submittal shall be as follows:

Cable groups (see table VII)	
1, 2, 3, 8, 10, 14	15 feet.
4, 6, 7, 11, 12, 15, 16	20 feet.
9	25 feet.
5 and 13	50 feet.

4.6.2.3 Marking of specimens. All test specimens shall be marked with the following information:

(a) Name of manufacturer.
(b) Cable designation (type and size).
(c) Manufacturing lot number or order number, either of the command or agency concerned, or the shipbuilder (if submitted for delivery).
(d) Footage represented by specimen.
(e) The coil or reel number from which the specimen came.
(f) Qualification reference.

4.6.2.4 Packaging of specimens for shipment. To facilitate handling, specimens may be shipped in boxes or wrapped coils.

4.6.3 Action in case of failure. If any specimen fails one or more of the specified verification or comparison inspection tests, the lot represented by that specimen shall be held in abeyance. The manufacturer shall have the option of resubmitting the lot, but only after the lot has been re-evaluated, the cause of failure determined, and corrective action has been taken. If the manufacturer resubmits the lot, the specimen representing the lot shall be subjected to the test or tests under which failure occurred plus any other tests which the qualifying activity may deem necessary to assure compliance with this specification. If the manufacturer chooses not to resubmit the lot, or if the lot is resubmitted and a failure is found, the lot shall be rejected. Once comparison inspection reveals a failure in a particular type and size of cable, all lots of the particular type and size subsequently produced shall be submitted for comparison inspection. After two successive lots pass comparison inspection, the manufacturer shall resume the normal rate of submission of samples for comparison inspection.

4.6.4 Action in case of rejection. Any completed cable which has been rejected and is not acceptable to the Government shall be recorded with the Government inspector. The record shall be made within 10 days after rejection, and shall show the type, size, lengths, and serial numbers of the marker tape recorded from each end of each length.

4.6.5 Test reports. One copy of laboratory reports covering comparison inspection shall be forwarded to the qualifying activity.

4.7 Conductor inspection. Conductors shall be tested on a lot basis for tensile strength, elongation, and coating, as applicable. The weight of a lot shall be not more than 1000 pounds. Records of tests shall be kept and copies of the applicable records shall become a part of the test reports for the finished cable.

4.8 Test methods (physical).

4.8.1 Visual and dimensional examination. Each of the finished samples selected shall be visually and dimensionally examined, the materials examined, and the manufacturing process evaluated to determine that the completed cable conforms to this specification.

4.8.1.1 Material examination. Examination shall be made of conductors, types of insulation, types of jackets, and other component materials as well as limiting dimensions of each layer or stage when applicable to the cable under consideration.

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4.8.1.2 Manufacturer, conductor, and group identification. Examination shall be made for manufacturer's identification and for the proper conductor and group coding.

4.8.1.3 Dimensions. The cable shall be measured to determine that the wall thickness, centering, circularity, and dimensional requirements are in accordance with this specification. For insulation or other components nominally less than 1/32 inch thick, measurement of thickness shall be made with a pingage dial micrometer by placing the specimen on the pin and taking the reading immediately after the movable member comes to rest on the specimen.

4.8.1.4 Insulation and jacket repairs. Cable shall be examined to determine that insulation and jacket repairs meet the criteria of 3.7.

4.8.1.5 End seals. End seals shall be examined on both ends of each shipping length to insure that they have been properly applied.

4.8.2 Physical tests on insulation and jacket. Tests shall be made on insulation and jacket specimens which have been removed from completed cable to determine the physical properties of the materials before and after aging. The tests shall be made on the aged and unaged specimens at the same time insofar as possible.

4.8.2.1 Specimens. In the case of small conductors with thin walls, the specimen shall be the entire section of insulation. In the case of cable jackets or large conductors with thick walls, the specimen shall be either (a) a segment or section cut with a sharp knife held tengentially to the conductor or cable core or (b) a shaped specimen cut with a standard die.

4.8.2.1.1 Size. The cross-sectional area of specimens shall be as large as possible, but shall not exceed 0.030 square inch. All specimens shall be at least 6 inches long.

4.8.2.1.2 Calculation of area. The area of specimens shall be calculated before aging. Calculation of the area of the test specimens shall be made as follows:

(a) Where the total cross section of the insulation is used, the area shall be taken as the difference between the area of the circle whose diameter is the minimum average qutside diameter of the insulation and the area of the conductor. The area of a stranded conductor shall be figured from its maximum diameter. The following formula may be used to advantage:

$$A = 0.7854 (D + d) (D-d)$$

Where: A = Area of cross section in square inches.
D = Outside diameter of insulation in inches.
d = Diameter over copper in inches.

(b) Where a slice cut from the insulation by a knife held tangentially to the wire is used, and the slice so cut has the cross section of a segment of a circle, the area shall be calculated as that of the segment of a circle whose diameter is that of the insulation. The height of the segment is the wall of insulation on the side from which the slice is taken. The values may be obtained from a table giving the areas of segments of a unit circle for the ratio of the height of the segment to the diameter of the circle.

(c) Where the cross section of the slice is not a segment of a circle, the area shall be calculated from the cross measurement of the volume or from the specific gravity and the weight of a known length of the specimen having a uniform cross section.

(d) Where a portion of sector of circle has to be taken as where the conductor is large and the insulation thin, the area shall be calculated as the thickness times the width. This applies either to a straight test piece or one stamped out with a die, and assumes that corrugations have been removed by buffing.

(e) Where a portion of a sector of a circle has to be taken, as where the conductor is large and the insulation thick, the area shall be calculated as the proportional part of the area of the total cross section.

4.8.2.1.3 Temperature. Specimens for both the aged and unaged physical tests shall be at a temperature between 20° and 28°C., and shall have been at this temperature for at least 30 minutes prior to testing.

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4.8.2.1.4 Bench marks. All specimens shall be marked near the middle to provide a basis for determining elongation and set. The marks shall be 1 inch apart for polyethylene and 2 inches for all other materials. Aged specimens shall be marked after aging.

4.8.2.2 Apparatus. The testing machine shall be power-driven and preferably be the pendulum type. A spring-balanced type of apparatus may be used if provided with a device which will indicate the actual maximum load at which rupture takes place, and provision is made to prevent recoil of the spring. The testing machine shall be accurate within 1 percent of the breaking load.

4.8.2.3 Tensile strength and elongation. Test specimens having bench marks shall be placed in the jaws of the testing machine. The maximum distance between jaws shall be 4 inches for specimens with 2 inches between bench marks and a 2-inch distance between jaws for specimens with 1-inch spacing between bench marks. Each specimen shall be stretched at the rate of 20 inches per minute until rupture. The elongation shall be measured at the instant of rupture. The break shall be between the bench marks. The tensile strength shall be determined from the area of the unstretched and unaged specimen.

4.8.2.4 Set. The set test shall be made on unaged specimens having bench marks 2 inches apart. The specimen shall be placed in the jaws of the testing machine and then stretched at the rate of 20 inches per minute until the bench marks are 6 inches apart. The specimen shall be held in the stretched position for 5 seconds and then released. The distance between bench marks shall be measured 1 minute after release. The difference between this measurement and 2 inches shall be considered the set.

4.8.2.5 Accelerated aging procedures.

4.8.2.5.1 Conditioning of materials. The time and temperature for aging of specimens shall be as shown in table XII.

Table XII -- Accelerated aging of specimens.

Materials	Conditioning	Temperature °C.	Time (hours)
Insulations			
Polyethylene	Air oven	100	96
80°C. Polyvinylchloride	Air oven	100	96
105°C. Polyvinylchloride	Air oven	136	96
Cross-linked polyethylene	Air oven	120	168
	Air pressure heat	127	40
Ethylene propylene rubber (EPR)	Air oven	120	168
	Air pressure heat	127	40
Synthetic rubber	Air pressure heat	127	20
	Oxygen pressure	80	168
Jackets			
Special purpose thermoplastic	Hot oil	120	18
	Air pressure heat	126	20
Standard thermoplastic	Hot oil immersion	120	18
Polychloroprene	Hot oil immersion	120	18
	Air pressure heat	126	20
	Oxygen pressure	80	168
Chlorosulfonated polyethylene (CSP)	Hot oil immersion	120	18
	Air pressure heat	126	20
	Oxygen pressure	80	168

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4.8.2.5.2 Air oven. The test specimens shall be heated at the required temperature for the specific length of time (see table XII) in an oven which has forced circulation of fresh air. The oven temperature shall be maintained within ± 2°C. of the specified temperature.

4.8.2.5.3 Air pressure heat. The specimen shall be suspended in a pressure chamber. The pressure within the chamber shall be held to 80 ± 2 p.s.i. for the specified time and temperature. The temperature regulation for the test chamber shall be such that the required temperature within the chamber will be reached within 15 minutes after inserting the specimen, and will maintain the required temperature within ± 2°C. for the specified time (see table XII). At the end of the aging period the pressure shall be reduced to atmospheric pressure at a uniform rate over a 5-minute period.

4.8.2.5.4 Oxygen pressure. The test specimens shall be heated in an atmosphere of oxygen at a pressure of 300 ± 10 p.s.i. for the specified time and temperature (see table XII). The bomb temperature shall be recorded automatically on a chart, and controlled to ± 1°C. for the specified time. At the end of the aging period, the bomb pressure shall be reduced at a uniform rate requiring at least 5 minutes to reach atmospheric pressure, at which time the specimens shall be removed.

4.8.2.5.5 Hot oil immersion. Specimens shall be immersed in lubricating oil conforming to Symbol No. 2190 TEP of MIL-L-17331, at a temperature of 120 ± 2°C. for 18 hours. The specimens shall then be removed, blotted lightly to remove excess oil, then suspended in air at room temperature for not less than 3-1/2 nor more than 4-1/2 hours before testing for tensile strength and elongation.

4.8.3 Tests on armor.

4.8.3.1 Physical. Specimens of the armoring wire shall be subjected to tests of tensile strength, elongation, springiness, and toughness before application to the cable.

4.8.3.1.1 Springiness. A length of armoring wire shall be coiled snugly around a 1/4-inch mandrel for several turns with a tension of not less than 100 grams nor more than 125 grams, and then released. The outside diameter of the coil, as measured in several places at or near its middle, shall not exceed 3/8 inch.

4.8.3.1.2 Toughness. The toughness of the wire shall be determined by bending a specimen back and forth, in a specially designed fixture, over a 0.030-inch radius support. Each 90-degree movement in either direction shall be taken as one bend. The bends shall be made under a uniform tension of 100 grams, and at a uniform rate not to exceed 50 bends per minute. The wire shall withstand a minimum of ten 90-degree bends before breaking.

4.8.4 Permanence of printing. The test is conducted on a horizontal abrasion machine using a length of No. 50-2/20 unbleached cotton braid, approximately 1/2 inch wide. This tape shall be drawn back and forth at right angles to the axis of the specimen and on a section of the specimen where the circuit identification is printed. The test setup is arranged so that an arc of contact of 135 degrees is established between the cloth tape and the surface of the wire. A stroke of 10 inches shall be used with a 1/2-pound weight suspended from one end of the tape and the other end fastened to the reciprocating table. The machine should be adjusted for a motion of about 28 cycles per minute (one cycle consisting of one back and forth motion). The tape shall not be used for more than 500 cycles per side.

4.8.5 Cold bend.

4.8.5.1 Specimens. Specimens shall consist of a 5-foot length of completed cable.

4.8.5.2 Procedure. The specimen shall be straightened and, if necessary, kept straight by securing to a straight wooden bar. The straight specimen shall be placed in a cold chamber and conditioned for a minimum of 6 hours at a temperature of 0°C., unless otherwise specified on the applicable specification sheet. The specimen shall then be removed from the refrigerator and, without delay, bent 180 degrees around the appropriate sized wooden mandrel (see applicable specification sheet) at a rate of approximately 20 degrees per second. The specimen shall be secured to maintain its bent position.

4.8.5.3 Observation. While the specimen is in the bent position, it shall be dissected and a visual examination made of all its component parts. When defects or disarrangements are discovered in any component, examination shall be made of specimens in the "as received" condition to determine whether the

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defect existed prior to the cold bend test. Any changes or derangements, such as torn tapes or broken or cracked components resulting from the test, of a nature or to an extent as to impair the performance of the cable, in service, shall be cause for rejection.

4.8.6 Cold working.

4.8.6.1 Specimen. Specimens shall consist of a 20-foot length of completed cable.

4.8.6.2 Procedure. The specimen shall be placed in a cold chamber and conditioned at minus 54°C. for a minimum time of 6 hours. Immediately after the conditioning, and while still in the cold chamber, the specimen shall be worked around two 12-inch diameter grooved sheaves in such a manner than an "S" bend of 180 degrees in each direction is imposed progressively on the entire length for a total of four times in each direction. The specimen shall be pulled over the sheaves at the rate of 50 ± 5 feet per minute.

4.8.6.3 Special conditioning (for qualification only). Prior to the cold working test for qualification, the specimen shall be preconditioned by soaking in salt water (approximately 3 percent salt by weight) at room temperature for 3 weeks and then, without delay, placed in the cold chamber. For this test the number of cycles for pulling the specimen over the sheaves shall be ten.

4.8.6.4 Observation. If, after the working, any defect or derangement is found in any component of the specimen, examination shall be made of a specimen in the "as received" condition to determine whether the defect existed prior to the bending. Any changes, broken or cracked components resulting from the cold working, of a nature or to an extent as to impair the performance of the cable in service, shall be cause for rejection.

4.8.7 Drip.

4.8.7.1 Specimen. The specimen shall consist of an 18-inch length of completed cable.

4.8.7.2 Procedure. Specimens shall be suspended vertically in an oven for 18 hours and at a temperature of 95 ± 1°C.

4.8.7.3 Observation. No material shall fall away from the lower end of the specimen during the test.

4.8.8 Thermoplastic flow.

4.8.8.1 Specimen. The specimen shall consist of a section of completed cable long enough to allow for two complete turns around the mandrel, securing the cable to the mandrel, and stripping of the cable ends.

4.8.8.2 Procedure. The ends of the specimen shall first be prepared for electrical connections as required herein. The specimen shall then be wrapped for two close turns around a mandrel having a diameter of 10 to 12 times the diameter of the specimen. The specimen shall be secured to the mandrel and placed in an oven. Electrical connections shall be made so that all conductors will be tested from conductor to shield at 70 volts, and also between conductors at 120 volts, when two or more conductors have a common shield. The specimen shall be kept in the oven for 6 hours at a temperature of 120 ± 2°C. during which time the voltage shall be applied to all conductors.

4.8.8.3 Observation. Electrical breakdown between conductors or from conductors to shield shall constitute a failure.

4.8.9 Watertightness.

4.8.9.1 Specimen. Specimens shall consist of 5-foot lengths of completed cable. If there is any question as to the satisfactory processing of the completed cable as to the starting and finishing ends, the manufacturer shall cut off and discard such ends before selecting the sample for test.

4.8.9.2 Procedure. One end of the specimen shall be placed in a terminal fitting which will allow water pressure to be applied directly to the exposed cross-sectional area of the end of the cable. Exposure of the sides of the cable to the water shall be kept to a minimum, and the fitting shall not exert radial compression against the cable. Unless otherwise approved by the qualifying activity, the sealer used for the packing gland in the terminal fitting shall be a metal alloy having a maximum melting point of 88°C. The specimen shall be subjected to a water pressure of 25 p.s.i. for a period of 6 hours.

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4.8.9.3 Observation. The acceptable amount of water leakage through the specimen shall be as shown in table XIII.

Table XIII — Limits for water leakage.

Total circular-mil area of condrs. in cable	Maximum water leakage in cubic inches (cu. in.) during 6-hour test at 25 p.s.i.			
	Full contract price	95 percent of contract price	90 percent of contract price	80 percent of contract price
0 to 9,000	1 cu. in.	2 cu. in.	3 cu. in.	4 cu. in.
9,050 to 15,000	2	3	4	5
15,050 to 25,000	3	4	5	6
25,050 to 50,000	4	5	6	8
50,050 to 100,000	5	6	7	9
100,050 to 200,000	6	7	8	10
200,050 to 500,000	7	8	9	11
500,050 to 800,000	8	9	10	12
800,050 to 2,000,000	9	10	11	13

Note 1. — Circular-mil area to be based on nominal circular-mil area required by the specification.
Note 2. — Cable with water leakage in excess of the values shown in the 80 percent column will not be acceptable to the Government.

4.8.10 Hydrostatic (open end).

4.8.10.1 Specimens. Specimens shall consist of a 5-foot length of completed cable.

4.8.10.2 Procedure. One end of the specimen shall be fitted into the appropriate size, double-ended bulkhead stuffing tube which shall conform to Drawing 815-F-1197030. Water pressure shall be applied to the stuffing tube end of the specimen. The pressure and duration of application shall be as required by the applicable specification sheet.

4.8.10.3 Observation. Unless otherwise specified by the specification sheet, no water leakage through the specimen will be permitted, and there shall be not more than 1/4-inch slippage of cable core or components.

4.8.11 Breaking strength.

4.8.11.1 Specimen. A specimen shall consist of a 6-foot length (approximately) of completed cable.

4.8.11.2 Apparatus. The testing machine shall be power-driven, and shall be equipped with a dial, scale, or automatic recorder that will record within ± 1 percent of the force required to break the specimen.

4.8.11.3 Procedure. One end of the specimen shall be wrapped two full turns around the upper mandrel and secured in place to provide sufficient snubbing action to prevent slipping. The other end of the specimen shall be secured in a similar manner to the lower mandrel. The distance between the jaws with the specimen in place shall be not less than 6 inches. The jaws shall be separated at a uniform rate of approximately 1 inch per minute until the specimen breaks.

4.8.11.4 Observation. The breaking strength of the specimen shall be not less than that required by the applicable specification sheet.

4.8.12 Bending endurance.

4.8.12.1 Specimen. The specimen shall consist of a 30-inch length of completed cable.

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4.8.12.2 Procedure. The specimen shall be conditioned to the temperature specified on the specification sheet and maintained at that temperature for the duration of the test. All conductors of the specimen shall then be connected to form a single series circuit which shall be electrified for the duration of the test for the purpose of detecting conductor breakage. The specimen shall be suspended loosely between two rollers, with the upper end of the specimen fastened to the bending apparatus. Each roller shall have a diameter approximately three times the diameter of the specimen. By means of a weight attached to its lower end, the specimen shall be maintained under a tension stress of 40 pounds per square inch ± 2 of its cross-sectional area. The device used to attach the weight shall apply uniform, radial compression to the core of the specimen. The specimen shall be bent at its midpoint, 90 degrees in each direction over the rollers at a rate of 12 to 14 complete cycles (360 degress total travel) per minute. After completion of the required number of bending cycles, the specimen shall be removed from the test apparatus and subjected to the dielectric strength test specified on the applicable specification sheet.

4.8.12.3 Observation. After completion of the dielectric strength test, the specimen shall be dissected and visually examined. Any evidence of dielectric breakdown, breaking of conductor strands, cracking of insulation material, or cracking of jacket material shall constitute failure.

4.8.13 Twisting.

4.8.13.1 Specimen. The specimen shall consist of one length of completed cable. The length shall be 24 inches for cables having a diameter of 5/8-inch or less, and 18 inches plus 10 times the cable diameter in length for cables having a diameter larger than 5/8 inch.

4.8.13.2 Procedures. The specimen shall be conditioned to the temperature specified on the applicable specification sheet and maintained at that temperature during the test. The twisting apparatus may be an assembly powered by a single drive and equipped with means for counting the number of cycles impressed upon the specimen. The specimen shall be clamped in cable grips and subjected to 180 degrees of twist in each direction (360 degrees total twist) at the rate of 12 to 14 complete cycles per minute. The distance between grips shall be 6 inches plus 10 times the maximum diameter of the specimen. The upper grip shall be oscillated by the twisting apparatus. The lower grip shall be free to move vertically, but restrained from oscillation, and shall have a weight attached which stresses the specimen in tension 40 p.s.i. ± 2 of its cross-sectional area. After the required number of cycles, the specimen shall be given the dielectric strength test applicable to the type of cable being tested.

4.8.13.3 Observation. After the twisting and dielectric strength tests, the specimen shall be visually examined. Any dielectric failure, breaking of the conductors, or cracking of the insulation or jacket shall constitute a failure.

4.8.14 Tension.

4.8.14.1 Specimen. The specimen shall consist of a 6-foot length of completed cable.

4.8.14.2 Procedure. The specimen shall be secured at each end with a woven cable grip similar to type HD of MIL-G-160. A tension load as required by the applicable specification sheet shall be applied to the specimen through the cable grips for 1 hour, and then released. After release of the tension, the specimen shall be given the dielectric strength test applicable to the type of cable being tested.

4.8.14.3 Observation. The specimen shall withstand the tension and dielectric strength tests.

4.8.15 Stress endurance.

4.8.15.1 Specimen. The specimen shall consist of a 7-foot length of completed cable.

4.8.15.2 Procedure. The specimen shall be drawn back and forth under alternate loads of 490 and 180 pounds tension through 90 degrees around a 3-inch diameter pulley with a total travel of 18 inches in one direction, for a total of 500 cycles. After the required number of cycles the tension of the specimen shall be released, and the specimen shall be subjected to the dielectric strength test applicable to the type of cable being tested.

4.8.15.3 Observation. After the endurance and dielectric strength tests, the specimen shall be visually examined. Any dielectric failure, breaking of conductors, or cracking of the insulation or jacket shall constitute a failure.

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4.8.16 Pressure cycling (qualification only).

4.8.16.1 Specimen. A specimen shall consist of a 50- to 70-foot length of completed cable.

4.8.16.2 Procedure. The specimen shall be installed in a water pressure chamber with both ends brought out approximately 1 foot through a double-ended bulkhead stuffing tube of appropriate size and conforming to Drawing 815-F-1197030. The protruding cable ends shall be prepared for capacitance and insulation resistance testing. The specimen shall be subjected to five pressure cycles at the pressure specified on the applicable specification sheet. The pressure shall be reduced to zero between cycles. The pressure for the first and fifth cycles shall be held for 30 minutes. The pressure for the second, third, and fourth cycles shall be held only long enough to make the electrical tests. Capacitance and insulation resistance measurements shall be made before and after the pressure cycling and just before the pressure is released at the end of each cycle.

4.8.16.3 Observation. The insulation resistance and capacitance values obtained throughout the test shall meet those required by the applicable specification sheet.

4.8.17 Flammability.

4.8.17.1 Specimens. Specimens shall consist of completed wire or cable approximately 18 inches in length.

4.8.17.2 Apparatus. The apparatus shall consist of an enclosure, supports, heater coil, spark plugs, flame travel gage, and associated accessories.

4.8.17.2.1 Enclosure. An enclosure of sufficient size to contain the specimen, supports, heater coil, spark plugs, flame travel gage, and their associated accessories shall be arranged to eliminate air drafts and permit clear view of the interior through shatterproof glass windows. Vent holes, distributed around the sides adjacent to the base, shall be provided to admit fresh air when an exhaust fan connected to the top of the enclosure is operated at a minimum suction just sufficient to carry off smoke and gases.

4.8.17.2.2 Supports. Supports shall be suitable for holding the specimens in a vertical position with an unsupported span not less than 14 inches in length. The lower end of the specimen shall be wrapped with varnished cambric in such a manner that gases released through this end are diverted toward the spark plug.

4.8.17.2.3 Heater coil. Heater coils shall consist of seven turns of 0.102-inch diameter resistance wire conforming to composition E of QQ-R-175, space wound to 0.25-inch between centers. The inside diameter of the coil shall exceed the overall diameter of the specimen by not less than 0.5 inch nor more than 0.6 inch. The lower end of the heater coil shall be located 1.5 inches above the top of the lower specimen support.

4.8.17.2.4 Spark plugs. Two spark plugs, with extended electrodes spaced 1/8-inch from the surface of the specimen and the plugs located on diametrically opposite sides of the specimen, shall be placed with their longitudinal centerlines in a horizontal plane 1/2-inch above the top of the heater coil, to ignite the gases emitted from the heated specimen. An electric circuit adequate to maintain continuous sparking at the electrodes during the specified time shall be provided. The spark plugs shall be mounted in such a manner that they may be moved away from the specimen after ignition takes place so as not to impede the travel of the flame and to prevent their electrodes from becoming fouled by soot.

4.8.17.2.5 Flame travel gage. A flame travel gage shall be positioned near the specimen to judge the distance of flame travel without appreciably impeding the progress of the flame.

4.8.17.3 Procedure. With the enclosure closed and ventilated, the specimen centered in the heater coil, and the spark plugs and flame gage properly located, a stop watch shall be started simultaneously with the energizing of the heater coil and spark plugs. A constant current as specified in table XIV shall be supplied from a transformer source to the heater coil. Ignition shall be considered as occurring when the flame transfers from the escaping gases to the surface of the specimen and continues there, disregarding the flashes which may occur in the gassing space prior to the sustained flame. The time required to ignite the specimen shall be recorded. For all armored cable, the heating current shall be turned off 30 seconds after ignition occurs. When armored cable does not ignite in 250 seconds or less, the specimen is considered satisfactory, and the test shall be discontinued. For all unarmored cables where ignition time is less than 60 seconds, the heating current shall be turned off 60 seconds after being turned on; where the

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ignition time is greater than 60 seconds, the heating current shall be turned off when ignition occurs. Immediately after ignition occurs, the electrical supply to the spark plugs shall be cut off, and the plugs shifted away from the flame. The maximum distance to which the flame travels along the surface of the specimen, measured from the top of the heater coil, before extinction shall be noted and recorded.

Table XIV — Current in heating coil.

Nominal diameter of cable (inch)	Current amperes a. c. (r. m. s.)	Nominal diameter of cable (inches)	Current amperes a. c. (r. m. s.)
0.050 to 0.099	45	0.80 to 0.89	53
.100 to .19	46	.90 to .99	54
.20 to .29	47	1.00 to 1.09	55
.30 to .39	48	1.10 to 1.39	56
.40 to .49	49	1.40 to 1.79	57
.50 to .59	50	1.80 to 2.29	58
.60 to .69	51	2.30 to 3.00	59
.70 to .79	52		

4.8.17.4 Observation. The number of seconds that the specimen continues to burn until the cessation of all flaming, after the current in the heater coil has been cut off, shall be recorded as the time required for self-extinction. If a specimen tested fails to comply with any of the limitations specified in table XV as to flammability upon the first trial, two additional specimens shall be tested, and the results of all three trials averaged together. Specimens not passing the flammability limits shall constitute a failure.

Table XV — Flammability limits.

Cable type	Ignition time seconds (minimum)	Burning time seconds (maximum)
With armor	60	60
Without armor (except MRI)	30	120
MRI	25	60

4.9 Test methods (electrical).

4.9.1 Conductor resistance. Electrical resistance shall be determined on all conductors of all lengths of completed cable. The test procedure shall be in accordance with ASTM B193. The measured resistance shall be not greater than the value permitted by the applicable specification sheet.

4.9.2 Dielectric strength. Dielectric strength tests shall be made on all lengths of completed cable.

4.9.2.1 Apparatus. The dielectric tests shall be made with alternating potential from a source of ample capacity, but in no case less than 5 kilovolt-ampere, having a frequency not greater than 100 cycles per second and a wave shape approximately a sine wave under all test conditions. The testing voltage may be measured by any method, satisfactory to the Defense Contract Administration Service representative, which gives a root-mean-square value. The preferable method is by means of a voltmeter connected to voltmeter coil in the high-tension winding of the testing transformer, or to a separate instrument transformer.

4.9.2.2 Procedure. The test voltages and application (conductor to ground, conductor to conductor, shield to shield, and so forth) shall be as required by the applicable specification sheet. The time of application for all dielectric strength tests shall be 1 minute. The initially applied voltage shall be not greater than 600 volts. The rate of increase shall be approximately uniform and not over 100 percent in 10 seconds,

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nor less than 100 percent in 60 seconds. All unarmored cables requiring electrical tests for the jacket shall be immersed in a grounded water bath for a least 1 hour, and tested while still immersed, using the water as the ground.

4.9.2.3 Observation. All cable shall withstand without failure the dielectric tests specified on the applicable specification sheet.

4.9.3 Insulation resistance. The instulation resistance shall be determined for all conductor insulation and for cable jackets, when required by the specification sheet. The test shall be performed on each length of complete cable, immediately following the dielectric strength test.

4.9.3.1 Procedure. The leakage current shall be measured after 1-minute electrification with a direct current potential of not less than 200 nor more than 500 volts. Multi-conductor cable with unshielded conductors shall be tested between each conductor and all other conductors and to overall shield or armor, if present. Cable having shielded groups, such as pairs and triads, shall be tested between each conductor and all other conductors in the group and to the group shield. Cable with individually shielded conductors shall be tested between conductor and shield. For cable having combinations of shielded or unshielded, groups or single, the applicable procedure shall be used. Where cable jackets have insulation resistance requirements, the test shall be made between the overall cable shield and a water bath. The conductor or shield whose insulation is under test shall be connected to the negative terminal of the test equipment and readings shall be taken after 1-minute electrification.

4.9.3.2 Observation. The insulation resistance values at 15.5°C. shall be not less than required by the specification sheet. If the measurement is made at a temperature other than 15.5°C., the manufacturer shall correct the measured value to 15.5°C. If the insulation resistance is equal to or greater than that required when the measurement is made at a temperature greater than 15.5°C., no correction factor need be employed. The manufacturer shall demonstrate that the correction factor used is accurate for his compound.

4.9.4 Capacitance. When required by the applicable specification sheet, capacitance shall be determined for cables having shielded singles, shielded pairs, shielded triads, unshielded pairs, or unshielded triads.

4.9.4.1 Specimen. Specimens shall consist of a 10-foot, 2-inch length of completed cable with all shields removed for a distance of 1 inch from each end and the insulation removed for a distance of 1/2 inch from each end of all conductors. The length of the specimen shall be the shielded length.

4.9.4.2 Test equipment. A Q-meter and oscillator equipped with the external balance circuit, as shown in figure 1, is recommended. Other equipment may be used where it can be demonstrated that such an alternate will yield equally accurate results.

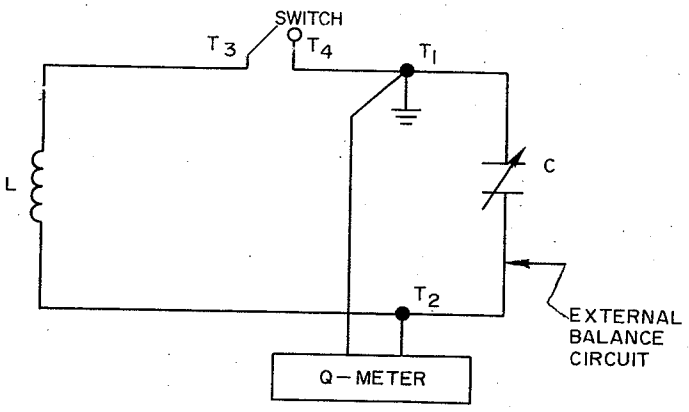


Figure 1 - Test equipment for capacitance test.

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L: Coil, of inductance suitable for the magnitude of the reactances to be measured.

C: Variable calibrated precision condenser.

T_1 and T_2: Terminals of Q-meter-T_1 grounded.

T_3 and T_4: Terminals in the inductive branch of the balance circuit. (T_4 can be identical with terminal T_1.)

S: Means of connecting T_3 to T_4.

The leads from L and C to the Q-meter terminals shall be as short as practicable. The test specimen or leads from the specimen shall be terminated at or as near as possible to the terminals of L.

4.9.4.3 Procedure (shielded singles). The shield at each end of the conductor under test shall be grounded. Measurements shall be made for each shielded conductor using the frequency required by the applicable specification sheet. Proceed as follows for each conductor of the specimen:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Connect the conductor to terminal T_2 and the shield to terminal T_1. Balance the circuit by adjusting C to a new value. Designate this value as C_1.

4.9.4.3.1 Calculation. The capacitance (C) per foot, for each conductor, shall be determined by the formula:

$$C = \frac{(C_0 - C_1)}{(\text{length of specimen in feet})}$$

4.9.4.4 Procedure (shielded pairs). All shields at each end of the specimen shall be grounded. The capacitance shall be measured for each pair at the frequency specified on the specification sheet. For each pair one conductor shall be designated as No. 1, the other conductor as No. 2 and the shield as No. 3. Proceed as follows for each pair:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Connect No. 1 and No. 3 to terminal T_1, and connect No. 2 to terminal T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance ($C_0 - C_1$) measured between terminals T_1 and T_2 as C_a.
(d) Disconnect No. 1 from terminal T_1 and No. 2 from terminal T_2. Connect No. 2 with No. 3 to terminal T_1 and connect No. 1 to terminal T_2. Balance the circuit by adjusting C to a new value C_2. Designate the capacitance ($C_0 - C_2$) as C_b.
(e) Disconnect No. 2 from terminal T_1 and connect it to terminal T_2. Balance the circuit by adjusting C to a new value C_3. Designate the capacitance ($C_0 - C_3$) as C_c.

4.9.4.4.1 Calculation.

(a) The mutual capacitance (C_m) per foot for each pair shall be determined by the formula:

$$C_m = \frac{2(C_a + C_b) - C_c}{4x(\text{length of specimen in feet})}$$

(b) The capacitance unbalance or coefficient of asymmetry (C_u) of a shielded pair, expressed in percent, shall be determined by the formula:

$$Cu = \frac{400\ (C_a - C_b)}{2\ (C_a - C_b) - C_c}$$

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4.9.4.5 Procedure (shielded triads). The shield at each end of the triad under test shall be grounded. Measurements shall be made for each triad using the frequency required by the applicable specification sheet. For each triad, one conductor shall be designated as No. 1, the second conductor as No. 2; the third conductor as No. 3, and the shield as No. 4. Proceed as follows for each triad:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Connect No. 2, No. 3 and No. 4 to terminal T_1. Connect No. 1 to terminal T_2. Balance the circuit by adjusting C to a new value C_1. Designate the capacitance $(C_0 - C_1)$ measured between terminals T_1 as C_a.
(d) Disconnect No. 2 from terminal T_1 and disconnect No. 1 from terminal T_2. Connect No. 2 to terminal T_2. Connect No. 1 with No. 3 and No. 4 to terminal T_1. Balance the circuit by adjusting C to a new value C_2. Designate this capacitance $(C_0 - C_2)$ as C_b.
(e) Disconnect No. 3 from terminal T_1 and disconnect No. 2 from terminal T_2. Connect No. 3 to terminal T_2. Connect No. 1 with No. 2 and No. 4 to terminal T_1. Balance the circuit by adjusting C to a new value C_3. Designate this capacitance $(C_0 - C_3)$ as C_c.
(f) Disconnect No. 1 and No. 2 from terminal T_1 and connect them with No. 3 to terminal T_2. Balance the circuit by adjusting C to a new value C_4. Designate this capacitance $(C_0 - C_4)$ as C_d.

4.9.4.5.1 Calculation.

(a) The mutual capacitance (C_m) per foot for each shielded triad shall be determined by the formula:

$$C_m = \frac{3(C_a + C_b + C_c) - C_d}{12 \times (\text{length of specimen in feet})}$$

(b) The capacitance unbalance or coefficient of asymmetry (K) of a shielded triad, expressed in percent, shall be determined by the formula:

$$K_1 = \frac{200\,(C_a - C_b)}{(C_a + C_b)}$$

$$K_2 = \frac{200\,(C_a - C_c)}{(C_a + C_c)}$$

$$K_3 = \frac{200\,(C_b - C_c)}{(C_b + C_c)}$$

Where: K_1 = The percent of capacitance unbalance of conductor No. 1 in relation to conductor No. 2.
K_2 = The percent of capacitance unbalance of conductor No. 2 in relation to conductor No. 3.
K_3 = The percent of capacitance unbalance of conductor No. 3 in relation to conductor No. 1.

Note: In the formula, the relative positions of C_a, C_b, and C_c may be transposed as necessary to avoid negative values.

4.9.4.6 Procedure (unshielded pairs). For unshielded pairs the procedure shall be the same as specified for shielded pair cables except that the overall shield, if any, and all conductors of the specimen, except the pair under test, shall be designated and treated as No. 3 when making measurements.

4.9.4.6.1 Calculations. The mutual capacitance (C_m) for each pair shall be determined by the formula as specified for shielded pairs.

4.9.4.7 Procedure (unshielded triads). For unshielded triads the procedure shall be the same as for triads except that the overall shield, if any, and all conductors of the specimen outside of the triad under test, shall be designated and treated as No. 4 when making measurements.

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4.9.4.7.1 Calculations. The mutual capacitance (C_m) and the capacitance unbalance (K) shall be determined by the formulas specified for shielded triads.

4.9.5 Characteristic impedance. The characteristic impedance of shielded singles or paired conductors shall be determined from the measured values of capacitance and inductance.

4.9.5.1 Specimen. The specimen shall consist of a 10-foot, 2-inch length of completed cable with all shields removed for a distance of 1 inch from each end, and the insulation removed for a distance of 1/2 inch from each end of all conductors. The length of the specimen shall be the shielded length.

4.9.5.2 Procedure. Measurement of the capacitance shall be made first and shall be in accordance with the procedure specified for the cable construction being tested. Measurements for determining the inductance shall be made as follows and on the same specimen used for determining the capacitance (see 4.9.4).

4.9.5.2.1 Measurements for shielded pairs shall be as follows:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading of the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Disconnect terminal T_3 from T_4. Connect the two conductors of the pair under test together at the far end of the specimen. Connect conductor No. 1 to terminal T_3. Connect conductor No. 2 and the shield of the pair to terminal T_4. Balance the circuit by adjusting C to a new value C_1. Designate the change ($C_0 - C_1$) as dC.

4.9.5.2.2 Measurement for shielded singles shall be made as follows:

(a) Connect terminal T_3 to T_4.
(b) Balance the circuit to resonance (maximum reading on the Q-meter) by adjusting C. Designate this reading of C as C_0.
(c) Disconnect terminal T_3 from T_4. Connect the shield to the conductor under test at the far end of the specimen. Connect the conductor to terminal T_3 and connect the shield to terminal T_4. Balance the circuit by adjusting C to a new value C_1. Designate the change ($C_0 - C_1$) as dC.

4.9.5.2.3 Calculations.

4.9.5.2.3.1 The inductance of the shielded pair or shielded single (Lp) may be determined by the formula:

$$Lp = \frac{dC}{W^2 C_0 (C_0 - dC)}$$

Where W = 2π times the frequency. If dC is 1 percent or less of C_0, the formula may be reduced to:

$$L_p = \frac{dC}{W^2 C_0^2}$$

4.9.2.3.2 The characteristic impedance of the shielded pair or shielded single (Z_0) may be determined by the formula:

$$Z_0 = \sqrt{\frac{L_p}{C_p}}$$

4.9.6 Pulse response time. The following procedure will be used for the determination of the pulse response times:

(a) The test will be performed on the reel length of a balanced cable circuit. The balanced condition can be achieved by several means:
  (1) Employing balanced test equipment.
  (2) Transformer - These must be good quality pulse transformers.
  (3) Electronic balancing circuits or devices.

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A dual beam oscilloscope of the fast rise type with a good band pass capability combined with a good quality square wave pulser connected as shown in the following schematic in figure 2 (simplified representative schematic).

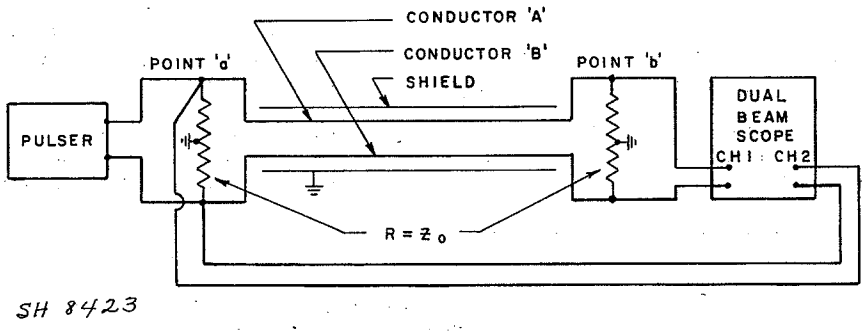


Figure 2 - Pulse response circuit diagram.

(b) It is important that the generator output pulse width be of sufficient duration to permit trace "b" of figure 3 to reach its full amplitude before "cutoff" is reached. For this cable a pulse duration of 2.0 microseconds will be used.

(1) Rise time determination. The dual beam oscilloscope should show a trace with the following characteristics:

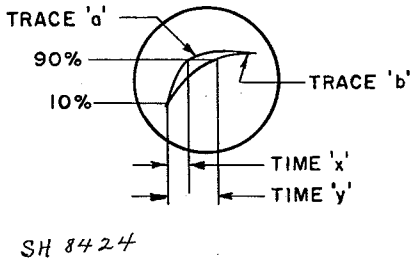


Figure 3 - Rise time.

Time "x" is the measurement of the arithmetic mean of the times of all components (pulser, scope, test leads to point "a", and so forth) connected to the input end of the cable. Time interval "y" is the mean of time "x" combined with the cable's rise time. It is important that the length, composition, and characteristics of the connecting leads to the oscilloscope from points "a" and "b" be identical, and as short as possible. Now, with both traces on the same time base, and coincident at the 10 percent amplitude value, determine the time duration of trace "b" from the 10 percent to 90 percent amplitude. Call this time "y". Next, insert these values ("x" and "y") into the formula $t_r = \sqrt{y^2 - x^2}$ where t_r is the cable's rise time, and "x" and "y" are the times previously determined.

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(c) Decay time may be determined as follows:

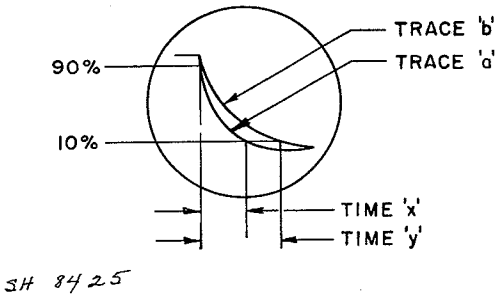


Figure 4 - Decay time.

Traces "a" and "b" are coincident at the 90 percent amplitude value. With the traces on the same time base, time interval "x" is the decay time mean as in the rise time determination, and time interval "y" is the mean of all decay times including that of the cable. As before, insert these values of "x" and "y" into the formula: $t_d - \sqrt{y^2 - x^2}$ where: t_d = the decay time of the cable: "x" and "y" are as previously described.

(d) If transformers or other balancing devices are used to provide the balanced condition, time "x" will have to be determined by reading the time of trace "b" in both cases with the cable out of the circuit, and the transformers or devices connected directly together, then time "y" will be determined as before with the cable inserted in the circuit. The difference between the traces "a" and "b" when the transformers or devices are connected directly together is the difference in the mean value of the rise and decay times caused by these items. One channel of the dual beam scope is connected at point (a) in this test to provide a reference for any possible changes to the input pulse caused by the insertion of the cable.

(e) Insert the values found for the rise and decay times into the formulas:

$$t_r/500 = \frac{500\ t_r}{l}$$

$$t_d/500 = \frac{500\ t_d}{l}$$

to determine the rise or decay time for a 500-ft. length of cable.

Where: t_r = rise time determined for sample,
t_d = decay time determined for sample,
l = sample length in feet

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4.9.7 Attenuation. Attenuation per unit length is defined as the logarithmic decrement in transmitted power. The attenuation, expressed in db per 100 feet, shall be measured at a sufficiently low-power level that the resulting temperature rise will be negligible. An acceptable method for measuring attenuation is as follows:

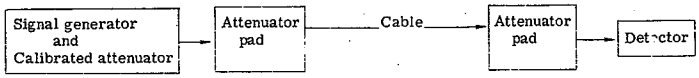


Figure 5 - Attenuation measurement.

In the block diagram (figure 5) a length of cable with an attenuation of at least 3 db shall be inserted between the connectors. The signal generator and calibrated attenuator shall be adjusted to produce a reasonable indication at the detector when the detector is tuned. The detector reading is noted, and the calibrated attenuator output level is recorded. The cable under test is then withdrawn and the circuit completed with the connectors (or a very short length of cable). With the detector tuned, the calibrated attenuator shall be readjusted to reproduce the original reading at the detector, and the attenuator output level is again recorded. Attenuation is then computed as follows:

$$A = \frac{100}{L} \text{ (difference in calibrated attenuator readings in db)}$$

Where:
A = Attenuation in db per 100 feet.
L = Length of cable under test in feet.

For measurements at frequencies of 400 mc. or less, the characteristic impedance of the attenuator pads and connectors shall preferably be the same as that of the cable under test.

4.9.8 Mutual inductance.

4.9.8.1 Specimen. The specimen shall consist of a 10-foot length of completed cable with the insulation removed for a distance of 1/2 inch from each end of all conductors. The braided copper shield shall be removed or pushed back to expose 1-1/2 inches of the twisted pair at both ends. The conductors of one end of each shielded pair shall be shorted.

4.9.8.2 Procedure. An approximate 1000-Hertz current shall be circulated in the red and green conductors, then in the red and blue conductors, then in the green and blue conductors for MCOS-5, and in the yellow and green conductor for MCOS-6. The induced voltage appearing across the open end of each shielded pair shall be measured to an accuracy of 1 microvolt while the current is flowing.

4.9.8.3 Calculations. The mutual inductance in henries shall be determined as follows:

$$\text{Mutual Inductance} = \frac{\text{Induced Voltage (microvolts)}}{2\pi \times \text{frequency (Hertz)} \times \text{current (amps)}}$$

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements in referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 All cables covered by this specification shall be shipped on reels, unless otherwise specified on the applicable specification sheet.

5.1.1 Reels. Reels shall be of a design, strongly constructed, with flanges of sufficient diameter to protect the cable from injury during handling and storage. Where the gross weight of cable and reel exceed 1000 pounds, the reels shall have metal hubs. The diameter of the drum shall be not less than fourteen times the overall diameter of the cable.

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5.1.1.1 Placement of cable on the reel. The cable on each reel shall have both ends readily available for testing without re-reeling. The inner end shall extend through a hole in the flange near the drum, and the outer end shall be available at the top of the reel. Both ends of the cable shall be sealed. Both ends of the cable shall be secured to the reel flange with 1/4 inch rope, applied with two half-hitches, one 6 inches and the other 4 inches (approximately) back of the end seal, then twisting the rope ends together, knotting them, pulling the cable up tight and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange. The inner end which extends through the flange hold shall be protected by a metal end cover.

5.1.1.2 Lagging. The cable wound on reels shall be covered with flexible waterproof barrier-material secured to the flange rims. The material used shall conform to UU-P-271 or 0.004-inch thick polyethylene film conforming to L-P-378. Reels shall be completely enclosed with wooden lagging with boards touching each other. Lagging boards shall be same thickness as the outside wooden reel flanges but not thicker than 2-inch commercial lumber, and shall be securely nailed to the outside wooden flanges of the reel with cement-coated nails. On reels 36 inches in diameter and larger, lagging shall extend within 1/4 inch of the outside edges of the flanges. On reels smaller than 36 inches in diameter, lagging shall be flush with the outside edges of the flanges. Lagging of reels of cable weighing 1000 pounds or more shall be strapped with two 0.1620 ± 0.003-inch (8-gage) high-tensile rough wires or two 3/4- by 0.035-inch steel straps. Lagging of reels of cable weighing less than 1000 pounds shall be strapped with two 0.0915 ± 0.003-inch (13-gage) high-tensile round wires or two 5/8- by 0.020-inch steel straps. All straps shall be stapled at intervals of approximately 15 inches. Lagging on metal flanges shall be secured by notched lagging of adequate thickness over the flanges and by steel strapping. A minimum clearance of 2 inches, but not less than one cable diameter, shall be provided between the inner face of the lagging and the outer layer of cable.

5.1.1.3 Alternate to lagging. Reels smaller than 42 inches in diameter which contain less than 1000 pounds of cable, shall be provided with a cushioning inner layer of asphalt-impregnated, corrugated fiberboard, or the equivalent, wrapped directly over the cable, so as to cover it completely. There shall be an outer covering consisting either of two layers of weather-resistant veneer paper conforming to PPP-V-205, or of two layers of three-ply weather-resistant toxic-treated plywood. The outer covering shall be wrapped directly over the inner layer so as to cover it completely, and secured in place between the reel flanges by means of at least two 3/8- by 0.020-inch steel straps, applied circumferentially.

5.1.1.4 Marking. In addition to any special marking required herein or in the contract or order, reels shall be marked in accordance with MIL-STD-129.

5.1.1.5 Yearly marking. In addition to any other required marking, reels shall be marked with a keyed series of colors to indicate the year of manufacture. The cycle of colors shall be repeated every fifth year. The reel marking shall consist of a strip approximately 2 inches wide, colored to designate the particular year of manufacture as follows:

Year of manufacture			Identifying color
1966	1971	1976	Red
1967	1972	1977	Green
1968	1973	1978	Orange
1969	1974	1979	Blue
1970	1975	1980	White

5.1.1.5.1 Location of yearly marking. The colored stripe on reels shall be applied circumferentially over the lagging and midway between the flanges. The stripe shall consist of one coat of commercial quality outside paint of the appropriate color. In addition to the stripe, both flanges of the reel shall be stenciled with 4 inch high figures to show the year of manufacture.

5.1.1.6 Standard reel markings: Each reel shall be plainly marked on both flanges with the following information:

(a) Reel number.
(b) Type and size of cable.
(c) Cable footage.
(d) Contract or order number
(e) Contractor's name.

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(f) Manufacturer's name (if other than contractor).
(g) Gross weight.

5.1.2 Coils. Coils shall contain one continuous length of cable (standard length, see table XVI).

5.1.2.1 Marking. In addition to any special marking required herein or in the contract order, coils and containers shall be marked in accordance with MIL-STD-129.

5.1.2.2 Yearly marking. In addition to any other required marking, coils shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2 inches wide and colored for the particular year of manufacture. The cycle of colors shall be the same as those used for yearly marking of reels.

5.1.2.3 Identification. Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information:

(a) Type and size of cable.
(b) Cable footage.
(c) Contract or order number.
(d) Contractor's name.
(e) Manufacturer's name (if other than contractor).
(f) Gross weight.

5.2 Packing. Packing shall be level A, B, or C in accordance with MIL-C-12000, as specified (see 6.2); however, cable packaged on reels as specified herein shall require no further packing for domestic or overseas shipment and storage (all levels).

6. NOTES

6.1 Intended use. For Navy use, ratings, applications, and characteristics for cable covered by this specification are specified in Section 9620-5 of General Specifications for Ships.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of cable required with reference to applicable specification sheet.
(c) Quantity required.
(d) Level of packing required (see 5.2).

6.2.1 Quantity. The quantity of each type and size on a contract or order should be specified as an integral multiple of the unit ordering length shown on the applicable specification sheet.

6.2.2 Lengths. The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XVI. In order to compensate for handling and probable loss on issue, remnant lengths are subject to the price reductions given in table XVI. Scrap lengths are not acceptable. It is not intended that items being procured in exact lengths for a specific job or ship shall necessarily be shipped in lengths which are integral multiples of the unit ordering length or that price reductions shall apply in the event that exact footages or fractional lengths are required and so indicated in procurement of cable for other than stock purposes.

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Table XVI -- Cable lengths.

Type of package	Nominal length (feet)	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
		No price reduction (feet)	No price reduction (feet)	Price reduction		Not acceptable (feet)
				5 percent (feet)	10 percent (feet)	
Coil	500	500 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1500	1650 to 1350	1349 to 450	449 to 300	299 to 150	149 to 0
Reel	2000	2200 to 1800	1799 to 600	599 to 400	399 to 200	199 to 0
Reel	2500	2750 to 2250	2249 to 750	749 to 500	499 to 250	249 to 0
Reel	3000	3300 to 2700	2699 to 900	899 to 600	599 to 300	299 to 0

6.3 Special Clause. Except when small quantities are to be purchased, invitations for bid and contracts or orders should specify the following:

"In order to compensate for handling and probable loss on issue, remnant lengths shall be subject to the price reductions shown in table XVI. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable on the contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximately two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in any combination of random lengths."

6.4 Comparison inspection will normally require forty calendar days from the date the sample is sent to the Government-designated laboratory to the date the inspection results are received by the Government inspector.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-915. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (See 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 Replacement of types. Cable types which have been replaced by existing types are shown in table XVII.

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Table XVII -- Replaced cable types.

Replaced type and applicable specification	Existing type and applicable specification
DDGT (MIL-C-2681)	DSGA (MIL-C-2194)
MDGA (MIL-C-915A (SHIPS))	MDU (MIL-C-915B (NAVY) and MIL-C-915C)
MDGB (MIL-C-915A (SHIPS))	MSCA (MIL-C-2194)
MDGD (MIL-C-2681)	MSCA (MIL-C-915B (NAVY) and MIL-C-915C)
MDGL (MIL-C-915A (SHIPS))	MDY (MIL-C-915B (NAVY) and MIL-C-915C)
MDGT (MIL-C-2681)	MDY (MIL-C-915B (NAVY) and MIL-C-915C)
MDGY (MIL-C-2681)	MDY (MIL-C-915B (NAVY) and MIL-C-915C)
MHFF (MIL-C-915A (SHIPS))	MHOF (MIL-C-915B (NAVY) and MIL-C-915C)
MSS-6 (MIL-C-915A (SHIPS))	7SS (MIL-C-915B (NAVY) and MIL-C-915C)
PBJX (MIL-C-915A (SHIPS))	TCJA (MIL-C-2194)
PBTX (MIL-C-915A (SHIPS))	TCTA (MIL-C-2194)
SHFS (MIL-C-915A (SHIPS))	Type D with nylon jacket (MIL-W-16878)

6.7 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - MI, EL, AV
Navy - SH, EC
Air Force - 80

User activities:
Army - WC, MU, ME
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Preparing activity:
Navy - SH
(Project 6145-0227)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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